Probabilistic Machine Learning

Lecture 03: Generative Models and Naive Bayes

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Speaker notes

Three modules. Three kinds of classifier: a rule maps x to a class without any probability; a discriminative classifier models $P(y \mid x; \theta)$ only; a generative classifier models $P(y)$ and $p(x \mid y)$ and obtains the posterior by Bayes' theorem, and can generate data. Read as a sample, the training table cannot fill a full $p(x \mid y)$, whose cells multiply with every attribute (the curse of dimensionality), so the model must be restricted — by a simpler structure, by parametric families, or both. Naive Bayes: conditional independence of the attributes given the class, which reduces the number of parameters from $2^p - 1$ to p per class for p binary attributes; fitting is counting, and a count of zero vetoes all other evidence unless it is smoothed. Fitting the parameters: Bayes' theorem applied to a parameter gives a posterior over it — for yes/no data with a Beta prior, by adding counts (Lecture 03). On that posterior, counting was its peak, the maximum likelihood estimate (MLE); smoothing its mean under a flat prior; and a prior's peak is the MAP estimate. The posterior can be drawn, so the spread a point estimate discards is visible.

Three choices are independent of each other: what to model (generative or discriminative), how to fit its parameters (MLE, MAP or the whole posterior), and how to decide on a class. MLE and MAP apply equally to logistic regression in Lecture 04, which treats linear and logistic regression as models of a conditional distribution and compares logistic regression with naive Bayes. Fraud detection (base rate 1 in 1,000, binary red flags) is the running example.

Where We Left Off

Lecture 02: 100 students, a count table.

	PML	DB	total
fail	10	6	16
pass	30	54	84
total	40	60	100

$$P(\text{PML} \mid \text{fail}) = \frac{P(\text{fail} \mid \text{PML})\,P(\text{PML})}{P(\text{fail})} = \frac{10}{16}$$

One hypothesis, **one** piece of evidence: Bayes' theorem reads the table backwards.

Speaker notes

The count table of Lecture 02 is a joint distribution of two variables, course and result, given as counts. Bayes' theorem read it backwards: the prior $P(\mathrm{PML}) = 40/100$, the likelihood $P(\mathrm{fail} \mid \mathrm{PML}) = 10/40$, the evidence $P(\mathrm{fail}) = 16/100$, so $P(\mathrm{PML} \mid \mathrm{fail}) = 10/16 \approx 0.63$.

A generative classifier is exactly this computation with the course replaced by the class y and the exam result by an attribute vector $x = (x_1, \ldots, x_p)$: the class prior $P(y)$ and the class-conditional distribution $p(x \mid y)$ are estimated from a labelled table and combined by Bayes' theorem.

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> **One** hypothesis, **one** piece of evidence: Bayes' theorem reads the table backwards.

Now the hypothesis is a class — fraud or honest — and the evidence a whole transaction. What changes?

From One Attribute to Many

Speaker notes

The four questions are the extensions this lecture takes up. Several variables: $p(x \mid y)$ is a joint distribution over all attributes, with one number per combination of values ($2^{30} - 1$ per class for 30 binary flags). Dependence: the joint is not determined by its margins, so each dependence is a number of its own; naive Bayes assumes them away. Numbers: a continuous attribute has a density, not a probability per value, so it needs bins or a parametric family. Probabilities at all: a rule maps x to a class without any distribution (a discriminant function); logistic regression models $p(y \mid x)$ only (Lecture 04); the generative route models $p(x, y)$.

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Many variables at once. Amount, country, hour, device — how do they combine into one probability per class?

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Probabilities at all? A rule *if amount > 1k € and new device, then fraud* also classifies.

Today: Naive Bayes and Its Two Assumptions

A classifier built from Bayes' theorem needs two assumptions to become estimable — rows 1 and 2 of Lecture 01's table of shortcuts:

assumption	replaces	what is lost	module
Conditional independence	the joint $p(x \mid y)$ by $\prod_j p(x_j \mid y)$	the dependences between attributes	Naive Bayes
Point estimate (MLE, MAP)	the posterior over a parameter by its peak	how sure we are about the parameter	Fitting the Parameters

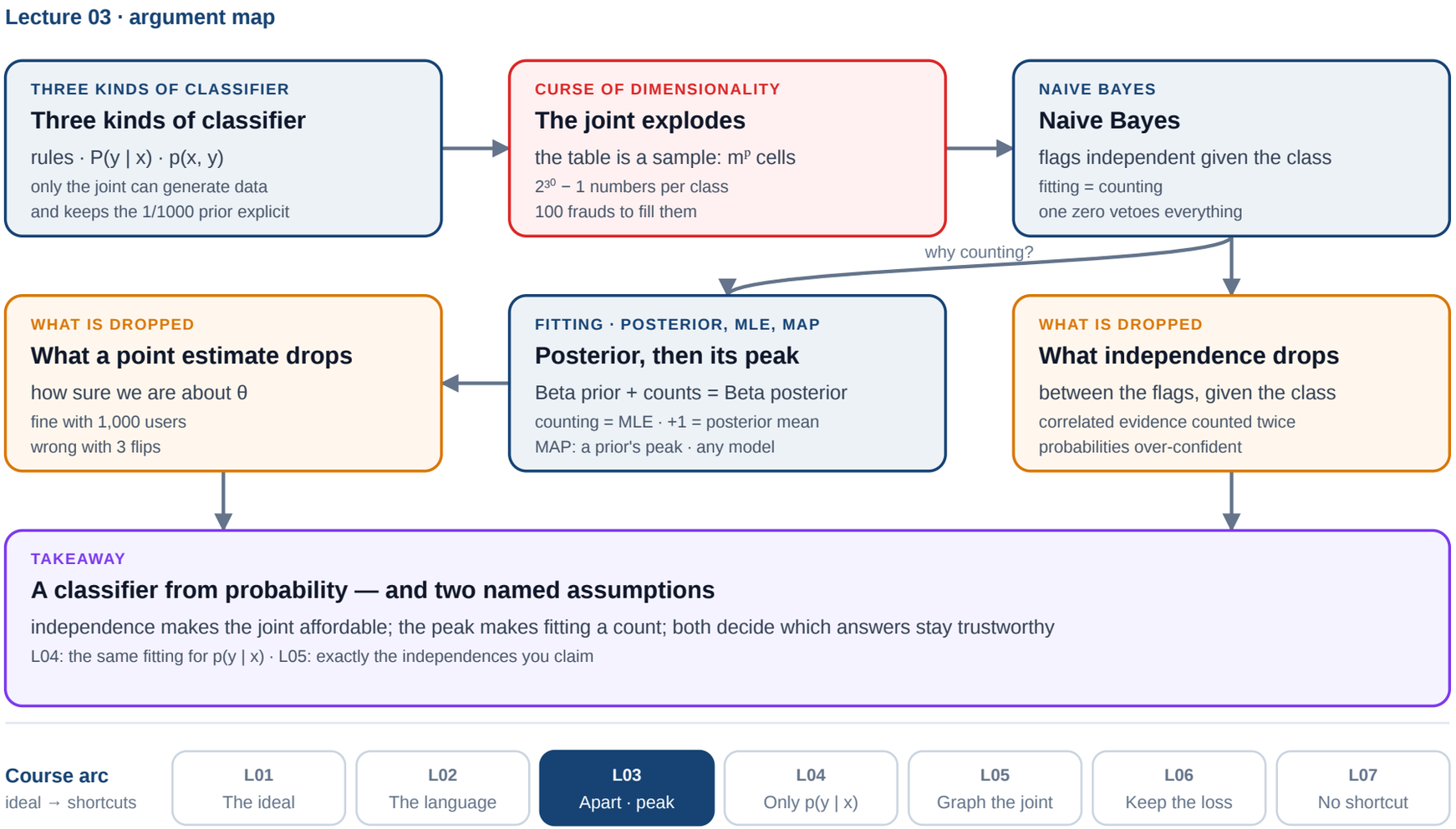
Speaker notes

The shortcut table of Lecture 01 lists the approximations the course makes to the ideal of carrying full distributions; there they are named "assume it apart" and "keep the peak". This lecture covers the first two, in the order a classifier needs them. Conditional independence replaces a joint distribution by a product of smaller factors and discards the dependences between the factors. A point estimate replaces the posterior over a parameter by its most probable value (MLE with a flat prior, MAP otherwise) and discards the uncertainty about that value.

Naive Bayes takes both: the joint $p(x \mid y)$ is replaced by a product of factors $p(x_j \mid y)$, and each factor is estimated by a relative frequency, which the last module identifies as a peak. A generative classifier as such requires neither shortcut: independence is what makes the joint affordable, and the peak is what turns fitting into counting. How the parameters are fitted is a choice of its own, made in the same way for discriminative models.

Running example: fraud detection with a base rate of 1 in 1,000 (prior odds 1 : 999) and a handful of binary red flags.

Argument Map



Speaker notes

The map follows the three modules. The generative route models $p(y)$ and $p(x \mid y)$ and converts with Bayes, keeping the 1/1,000 prior explicit; a discriminative model would fit $p(y \mid x)$ directly. Its problem is the joint: read as a sample, the table puts almost no data into each of the m^p cells ($2^{30} - 1$ numbers per class for 30 binary flags, against 100 frauds). Naive Bayes assumes the flags independent given the class, so fitting becomes counting; it drops the dependences between flags, so correlated evidence is counted twice and the probabilities are over-confident. A count of zero vetoes all other evidence, which motivates the last module: conditioning a parameter on the data gives a posterior; counting was its peak, the maximum likelihood estimate, +1 smoothing its mean, and MAP is a prior's peak. Keeping only a peak drops the spread, harmless with 1,000 observations and misleading with 3. How parameters are fitted is a separate choice from what is modelled: it applies unchanged to the discriminative models of L04. The course-arc strip places the lecture after the language of probability (L02) and before models of $p(y \mid x)$ only (L04) and graphical models (L05), which state exactly which independences are assumed.

Three Kinds of Classifier

Guiding Question: A rule, a model of $P(y \mid x)$, a model of how the data arise — what can each one tell us about a transaction, and what does the most informative one cost?

Data and Hypothesis as Random Variables

amount	distance	…	fraud?
49.90 €	3 km	…	no
1,250 €	900 km	…	**yes**
X_1	X_2	…	Y

- **Data** $x = (x_1, \ldots, x_p)$: the values of the attributes $X_1, \ldots, X_p$ — what we observe.
- **Hypothesis** y: the value of the class variable Y — what we want to know.
- **Decision:** the posterior $P(y \mid x)$.

Speaker notes

This fixes the notation for the rest of the course. Each column of a labelled table is a random variable, as in Lecture 02; the attribute columns are $X_1, \ldots, X_p$ and the label column is Y. A row is one realisation (x, y) of the vector $(X_1, \ldots, X_p, Y)$, and the table is a sample of n such draws.

The names follow the reading of Bayes' theorem: y is the hypothesis (which class), x the data (what was observed about the case). The count table of Lecture 02 has the course as the hypothesis, y, and the exam result as the data, x: $P(\mathrm{PML} \mid \mathrm{fail})$ is $P(y \mid x)$. The textbook letters vary (Bishop: $\mathbf{x}$ and t or $\mathcal{C}_k$; MacKay: D and H); in machine learning x and y are standard. Capital letters denote the random variables, small letters their values; x is a vector whenever there is more than one attribute.

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- **Decision:** the posterior $P(y \mid x)$.

The student table had hypothesis A = "PML" and evidence B = "fail". Which of the two is the data, which the hypothesis?

Parameters θ

A **model** is a family of distributions $p(\,\cdot\,;\theta)$. The **parameters** θ pick one member; training sets them.

Read the notation

$p(x \mid y;\theta)$ — the probability of data x, **given** the hypothesis y, **for** the parameter value θ.

The bar conditions on a random variable, the semicolon fixes a parameter.

Example

amount given fraud ~ normal: $\theta = (\mu, \sigma)$.

Fraud share $P(Y = \text{fraud};\theta) = \theta$: one number.

Speaker notes

Lecture 02 wrote a parametric family as $p(x \mid \theta)$. From here on the parameters are separated by a semicolon, $p(x; \theta)$ and $p(x \mid y; \theta)$, so that the bar is reserved for conditioning on random variables of the model (data and hypothesis). The two notations mean the same density; the semicolon says that θ is treated as a fixed, unknown number, not as a random variable.

The bar returns for θ once θ gets a distribution of its own: a prior $p(\theta)$ and a posterior $p(\theta \mid D)$ over the parameters given the training data D, as in Lecture 03. Every model in this course has parameters in this sense — the thresholds of a rule, the weights of logistic regression, the means and spreads of class distributions.

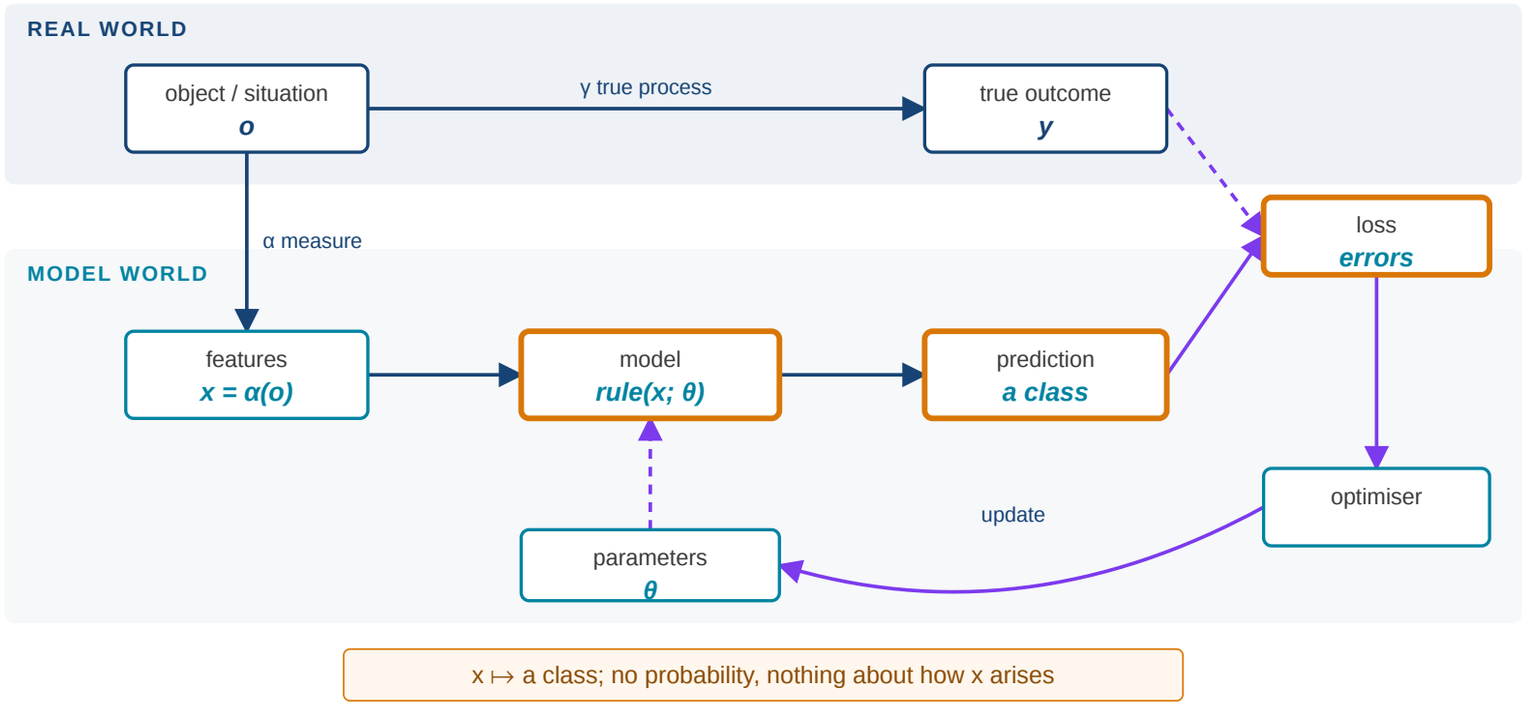
Rule-Based Classifier

A **rule-based classifier**: rules "if $C_k(x)$ then c_k" plus a default class. The **rules are the parameters** θ.

$x_1 > a,\ x_2 > b \rightarrow$ fraud, else honest: $\theta = (a, b)$, trained for the fewest errors

$$\hat{\theta} = \arg\min_{\theta} \sum_{i} 1[\hat{y}_i \neq y_i]$$

Discriminant function: rules



x ↦ a class; no probability, nothing about how x arises

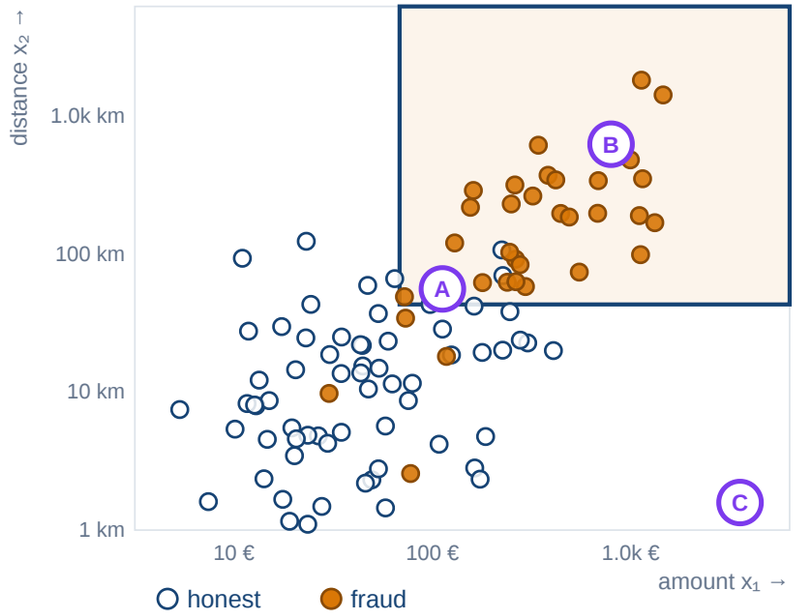
Speaker notes

The learning system of the introductory lecture with a rule learner in the model box. The features are the attributes x; the model computes $\mathrm{rule}(x;\theta)$; the prediction is a class; the loss counts errors on the training rows (the 0/1 loss, $1[\hat{y}_i \neq y_i]$ with $\hat{y}_i = \mathrm{rule}(x_i;\theta)$), and the optimiser searches over θ. The parameters of a rule set are its structure and its thresholds: which attributes the conditions test, the cut points a, b, the class of each rule and their order (first match wins). For a fixed form, as in the example, only the thresholds are free; a rule learner (RIPPER, CN2) or a decision tree (CART, C4.5) also chooses the conditions, greedily, because the number of possible rule sets grows combinatorially. A decision tree is a rule set whose conditions are the paths from the root to the leaves.

No component of the system represents a probability: there is nothing to condition on, nothing to marginalise and no prior to change. The 0/1 loss is piecewise constant in θ, so it is minimised by search over candidate thresholds, not by gradients.

Rules From Data

A and B get the same answer; C gets one without any warning



Learned rule

if amount > 68 €
and distance > 43 km
then fraud, else honest

training errors: 6 of 99

fraud share in the box: 94%
— the only "score" the rule has

A: fraud, 94%
just inside the box

B: fraud, 94%
deep inside the box

C: honest
5,000 € at 1 km — no such row

Which transaction is the rule surest about — A, B or C?

Speaker notes

The data are 105 labelled transactions with two attributes, the amount x_1 and the distance from the card holder's home x_2 (both on log axes); fraud is over-represented (a third) so that it is visible. The rule is learned from the table: among all rules of the form "amount $> a$ and distance $> b \rightarrow$ fraud", the thresholds (a, b) with the fewest training errors are chosen by trying every split between neighbouring values — the rule and the training criterion of the previous slide. The learned box has 6 training errors.

The rule assigns the same class to every point in its box. The only number it can attach is the share of fraud among the training rows in the box, the same for A (at the corner) and B (deep inside). C is a transaction unlike any row in the table; the rule classifies it as honest without any indication that the answer is unsupported.

What a Rule Can Not Say

- ✗ **How sure.** One share for the whole box — A and B alike.
- ✗ **Honestly.** The share is measured on the rows that chose the box: optimistic, 100% for a box searched until pure.
- ✗ **Under a new base rate.** The training mix is baked into the thresholds.
- ✗ **That an input is strange.** No model of x: C gets an answer like any other.

Examples

- **Fraud rule engines:** "amount > 1k € and new device → block".
- **Decision trees:** rules learned by splitting, one class per leaf.

A rule is a **discriminant function**: $x \mapsto$ a class. No distribution is modelled.

Speaker notes

A discriminant function maps an input directly to a class (Bishop, *Pattern Recognition and Machine Learning*, §1.5.4, distinguishes it from the two probabilistic approaches that follow). Rule sets and decision trees are the standard examples; a support vector machine is another.

The leaf or box share is a relative frequency $\hat{P}(y \mid x \in R)$ for the region R, not $P(y \mid x)$: it is constant over R, and because R was selected to make the share extreme, the training share overstates the share on new data (selection bias; the same effect makes a fully grown decision tree report 100% in every leaf). Calibrating such scores needs held-out data (a held-out set, not the training rows). Thresholds learned on a sample with one base rate encode that base rate; under a different one they are no longer optimal, and nothing in the rule says how to adjust them. Since no distribution over x is represented, the rule cannot measure how typical an input is or handle a missing attribute other than by ad hoc defaults.

The Joint Distribution $p(x_1, \ldots, x_p, y)$

A **probabilistic** classifier starts from the joint distribution of data and hypothesis. Every question is read off it:

question	what is read off the joint
Is this transaction fraud?	$P(y \mid x) = p(x, y) \,/\, p(x)$
How does fraud look?	$p(x \mid y) = p(x, y) \,/\, P(y)$
How common is fraud?	$P(y) = \sum_x p(x, y)$
How unusual is this transaction?	$p(x) = \sum_y p(x, y)$
The distance is missing — now?	sum x_2 out, then condition

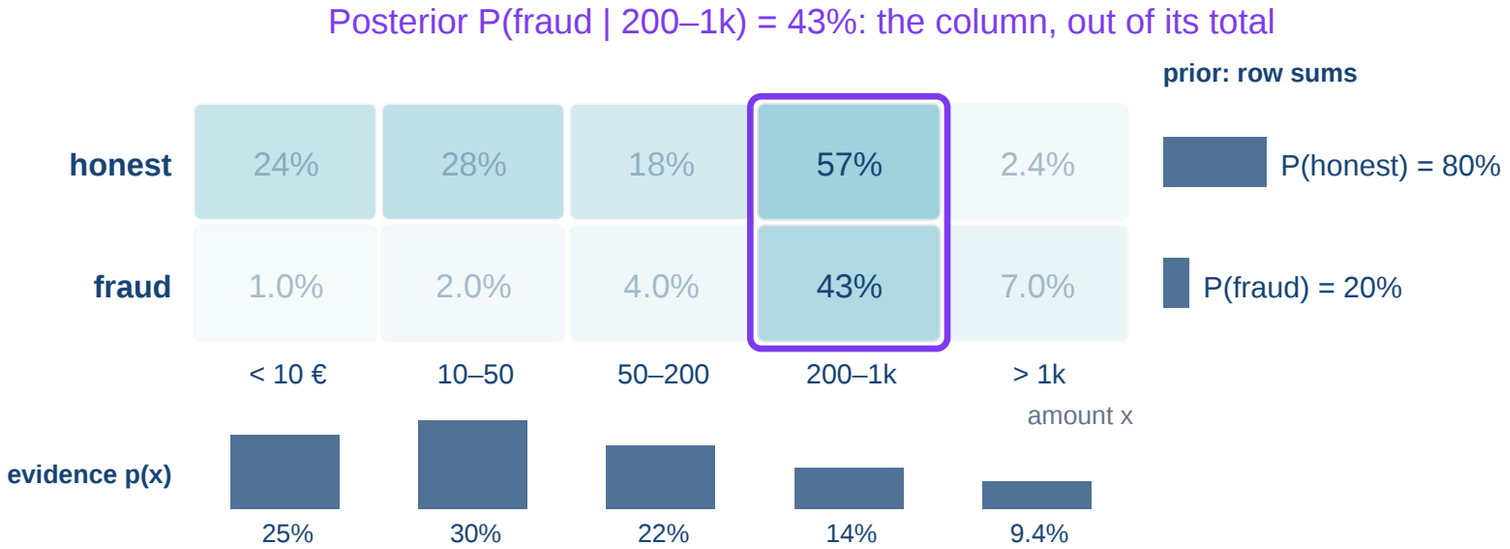
> The joint is the course's ideal: model everything, then condition.

Speaker notes

All quantities are obtained by the two operations of Lecture 02: conditioning (divide by a margin) and marginalisation (sum, or integrate, over the variables that are not asked about). For continuous attributes the sums over x are integrals and the probabilities densities.

The joint therefore contains strictly more than the decision needs. The two probabilistic classifiers differ in how much of it they model: the discriminative classifier only the conditional $P(y \mid x)$, the generative classifier the whole joint, factorised as $p(x \mid y)\, P(y)$. The cost of the whole joint is the subject of the rest of the module.

Discriminative: Model $P(y \mid x; \theta)$ Only



Fit **exactly** the decision: a distribution over y for every x.

- ✗ no $p(x)$, no $p(x \mid y)$
- ✗ no Bayes: nothing to turn around

Examples

Logistic regression · neural networks with a softmax output

Speaker notes

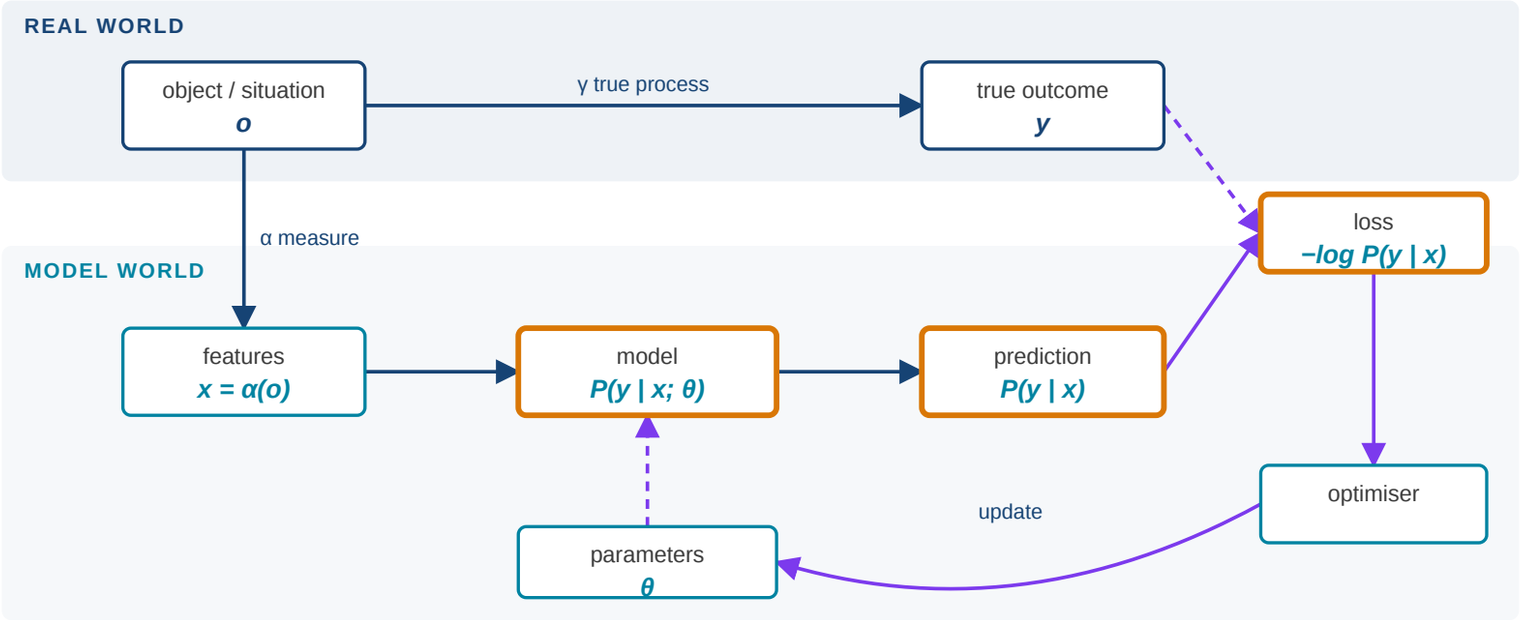
The heatmap is the joint $p(y, x)$ of the class and one attribute (the amount, five bins), with a fraud share of 20% so that the row is visible. Steps: the cells of the joint; the margins (prior $P(y)$ as row sums, evidence $p(x)$ as column sums); the fraud row against its total, the likelihood $p(x \mid \text{fraud})$; one column against its total, the posterior $P(y \mid x)$. A discriminative classifier models exactly the column shares: a family $P(y \mid x; \theta)$ whose parameters are fitted by maximising the probability of the training labels given the training attributes.

Unlike a rule, its output is a probability distribution over the classes, and its loss, $-\log P(y \mid x; \theta)$, rewards calibrated probabilities. Unlike a generative classifier, it never represents how x is distributed: the row margins and row shapes are not part of the model, so $p(x)$, $p(x \mid y)$ and the prior cannot be recovered, and Bayes' theorem cannot be applied to it. Logistic regression (Lecture 04) and neural networks with a softmax output layer are discriminative in this sense. A change of base rate must be corrected from outside (Lecture 04).

For 200–1k €: $p(\text{fraud}, x) = 0.2 \cdot 0.30 = 0.06$ and $p(\text{honest}, x) = 0.8 \cdot 0.10 = 0.08$, so $P(\text{fraud} \mid x) = 0.06/0.14 \approx 0.43$.

The Discriminative System in the Learning System

Discriminative model: p(y | x) only

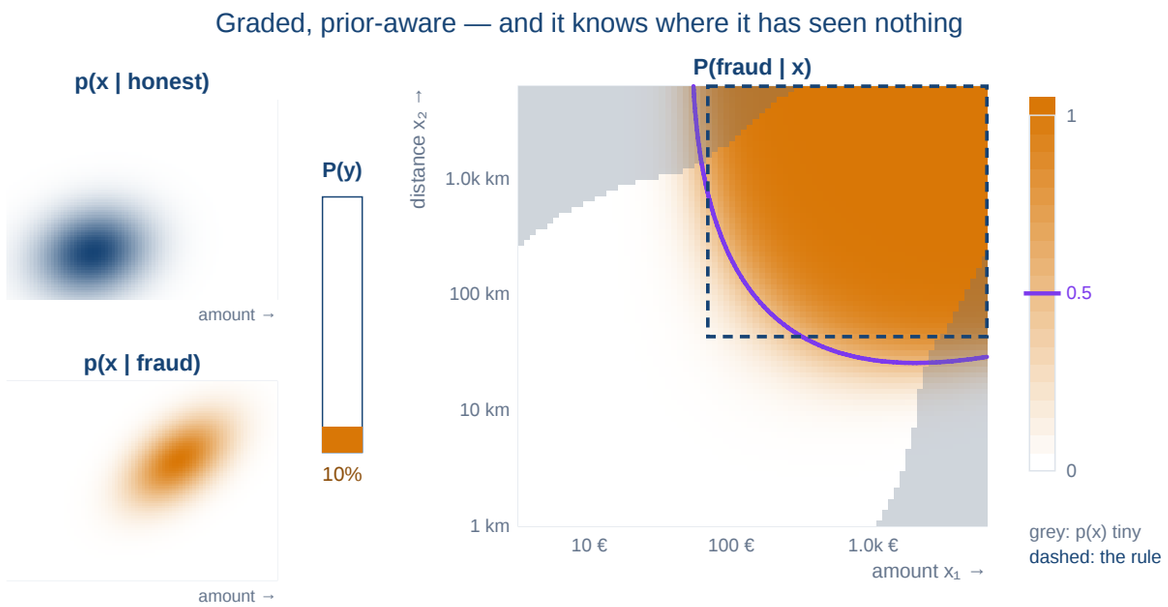


a distribution over y for each x; x itself is never modelled

Speaker notes

The model box computes $P(y \mid x; \theta)$; the prediction is a distribution over the classes, from which a class is chosen; the loss is the negative log-probability of the observed label, $-\log P(y \mid x; \theta)$, the cross-entropy. The features enter only as the condition: the model is evaluated at the observed x and makes no statement about which x occur. Fitting by maximum likelihood follows with logistic regression (Lecture 04).

Generative: Model $p(x \mid y; \theta)\, P(y; \theta)$



Set $P(\text{fraud})$ to 0.1%. What happens to the boundary — and why can the rule not do this?

Speaker notes

Each class has its own distribution over both attributes, here a two-dimensional normal in (log amount, log distance) with parameters $\theta_y = (\mu_y, \Sigma_y)$: two means, two spreads and a correlation per class. The prior $P(y;\theta)$ is one more parameter. Bayes' theorem combines them for every point, $P(\text{fraud} \mid x) = p(x \mid \text{fraud})P(\text{fraud}) \,/\, \sum_{y'} p(x \mid y')P(y')$, which the large panel shows; the violet line is where the posterior is 0.5.

Lowering the prior shrinks the fraud region: the likelihoods are unchanged, only the factor $P(\text{fraud})$ moves, so the model can be carried to a setting with a different base rate without refitting. At 1 in 1,000 the boundary retreats to where the likelihood ratio exceeds 999. Step 3 adds the rule of the previous slides (dashed) and greys out the region where $p(x)$ is tiny: there the posterior is still defined, but $p(x)$ says that no training data supports it. The sliders move the fraud mean and the correlation within the fraud class; the button draws transactions from the model.

Generating Data

Run the model forwards: draw $y \sim P(y)$, then $x \sim p(x \mid y)$ — a new data point.

Naive Bayes spam filter

Words drawn one by one from $p(\text{word} \mid \text{spam})$: *free winner click offer meeting now* — each word fits, the sentence does not.

Language models

$p(x_1, \ldots, x_T) = \prod_t p(x_t \mid x_{<t})$, fitted by maximum likelihood, sampled word by word.

Speaker notes

Sampling follows the factorisation $p(x, y) = p(x \mid y)P(y)$ from left to right: first the class, then the attributes given the class. Drawing a missing attribute from $p(x_j \mid x_{-j}, y)$, the conditional given the observed attributes, fills in part of a data point; a discriminative model has no distribution over x to draw from.

The naive Bayes example anticipates the next module: under the naive assumption each word is drawn independently given the class, so the words are individually typical of spam but do not form a sentence — the dependences between words are exactly what the assumption drops. An autoregressive language model keeps every dependence by the chain rule of probability, each factor $p(x_t \mid x_{<t})$ being one shared neural network evaluated on the context; it is fitted by maximum likelihood (cross-entropy on the next token). Whether a model can generate is therefore a property of *what* it models, not of *how* its parameters are fitted. Each factor of a language model, taken alone, is a classifier over the vocabulary; the product of all factors is the generative model.

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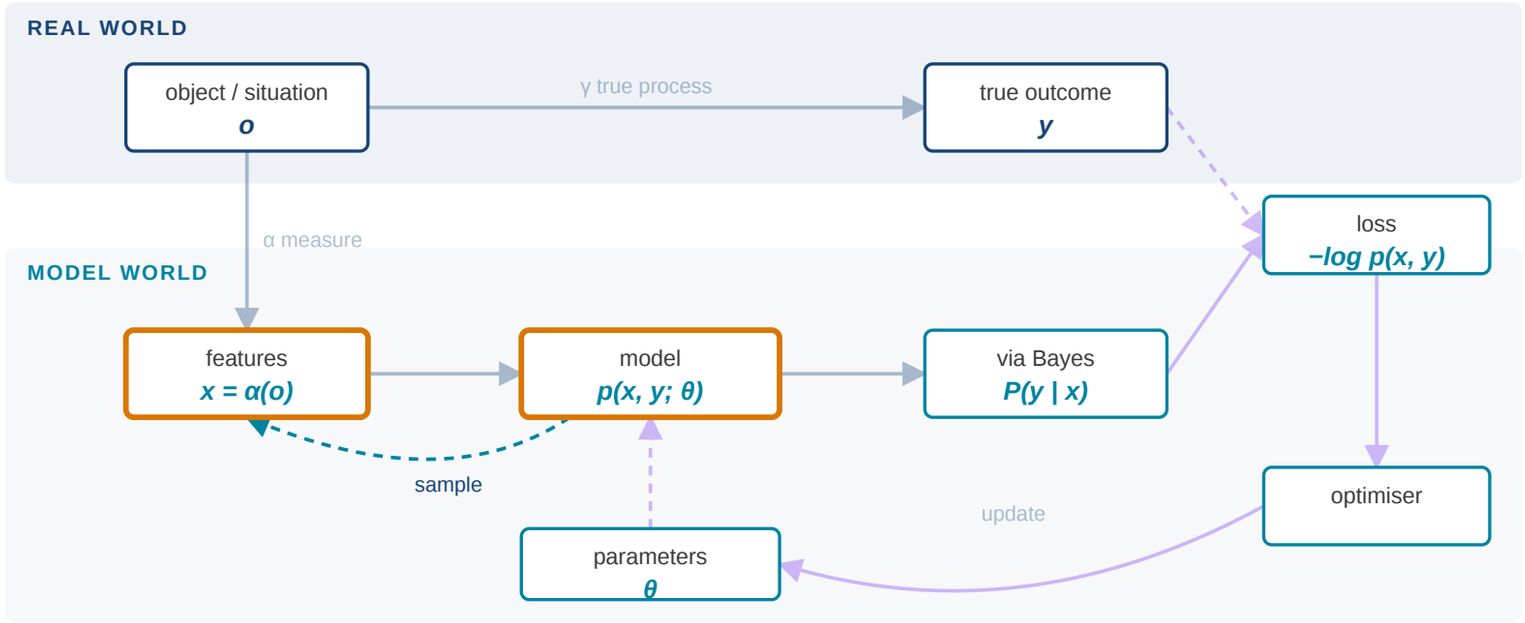
Language models

$p(x_1, \ldots, x_T) = \prod_t p(x_t \mid x_{<t})$, fitted by maximum likelihood, sampled word by word.

Neither rules nor $P(y \mid x)$ can do this: generating needs a model of x.

The Generative System in the Learning System

Generative model, sampling



draw y, then x from p(x | y): a new data point the model would produce

Speaker notes

The model box computes the joint $p(x, y; \theta)$; the prediction $P(y \mid x)$ follows by Bayes' theorem; the loss under maximum likelihood is $-\log p(x, y; \theta) = -\log p(x \mid y; \theta) - \log P(y; \theta)$, which scores the features as well as the label, so the features box is part of what is fitted. The second step runs the model without input: from the parameters to a sampled data point.

Three Kinds of Classifier

	rules	discriminative	generative
models	$x \mapsto$ class	$P(y \mid x; \theta)$	$p(x \mid y; \theta)\, P(y; \theta)$
how sure	✗	✓	✓
new base rate	✗ refit	correct from outside	✓ swap $P(y)$
unusual input	✗	✗	✓ $p(x)$ small
missing field	✗ ad hoc	✗ ad hoc	✓ sum it out
generate data	✗	✗	✓
examples	decision tree	logistic regression	naive Bayes, language model

The price of the right column: a distribution over **all** attributes, per class.

Speaker notes

New base rate: the posterior odds are the prior odds times the likelihood ratio, so a generative model only replaces $P(y)$, provided the class-conditional distributions are the same in both settings. A discriminative model absorbs the training base rate into its parameters; it can be corrected by multiplying its odds by the ratio of new to old prior odds, which requires knowing the training base rate. A rule has to be re-learned.

Missing attributes: $p(x_{\text{obs}} \mid y) = \sum_{x_{\text{mis}}} p(x_{\text{obs}}, x_{\text{mis}} \mid y)$. Without a model of x the alternatives are imputation or a separate model per pattern of missing fields.

Unusual inputs: $p(x) = \sum_y p(x \mid y) P(y)$ is small for an input unlike the training data (outlier, distribution shift); $P(y \mid x)$ is normalised for every x and cannot show this.

The comparison of accuracy, which favours the discriminative classifier with much data and the generative one with little, follows in Lecture 04.

Bayes' Theorem With Two Red Flags

Two flags at once, D = new device and F = foreign IP: the evidence is a **conjunction** of events.

$$P(\text{fraud} \mid D \cap F) = \frac{P(D \cap F \mid \text{fraud}) \cdot P(\text{fraud})}{\sum_y P(D \cap F \mid y)\, P(y)}$$

Among frauds, 60% show D and 70% show F. How many show both?

Speaker notes

This is Bayes' theorem for multiple events of Lecture 02 with the classes as the partition and the evidence $B = D \cap F$. Nothing in the theorem changes; the difficulty is entirely in the likelihood of the conjunction. (A conjunction on the other side, $P(E_i \cap E_k \mid B)$ for two classes, is zero: classes are disjoint. Several events enter a classifier as several pieces of evidence.)

The bounds: for probabilities $a = 0.60$ and $b = 0.70$ the intersection lies in $[\max(0, a + b - 1), \min(a, b)] = [0.30, 0.60]$ (Fréchet bounds). The lower end is the most "avoiding" overlap, the upper end has every D-fraud also showing F. The single-flag rates therefore do not determine $P(D \cap F \mid \text{fraud})$; it must be estimated separately. In the running example it is 0.55 among frauds and 0.04 among honest transactions, the figures used again in Lecture 03.

The general form: for flags $E_1, \ldots, E_p$, the likelihood $P(E_1 \cap \ldots \cap E_p \mid y)$ is needed for every combination of present and absent flags, 2^p patterns with $2^p - 1$ free numbers per class. The chain rule writes it exactly as $P(E_1 \mid y)\, P(E_2 \mid E_1, y) \cdots P(E_p \mid E_1, \ldots, E_{p-1}, y)$; each later factor conditions on more events, so it is estimated from fewer rows. Dropping the conditioning events in every factor gives the product $\prod_j P(E_j \mid y)$, which is exact iff the flags are conditionally independent given the class. With D and F the product gives 0.42 instead of 0.55 among frauds and $0.10 \cdot 0.05 = 0.005$ instead of 0.04 among honest transactions.

Bayes' Theorem With Two Red Flags

Two flags at once, D = new device and F = foreign IP: the evidence is a **conjunction** of events.

$$P(\text{fraud} \mid D \cap F) = \frac{P(D \cap F \mid \text{fraud}) \cdot P(\text{fraud})}{\sum_y P(D \cap F \mid y)\, P(y)}$$

Among frauds, 60% show D and 70% show F. How many show both?

1. **Not fixed by the parts.** Anything from 30% to 60% fits; the data say 55%. $P(D \cap F \mid y)$ is a number of its own.
2. **One number per combination.** p flags: $2^p - 1$ per class.
3. **Ever rarer.** Each added flag shrinks the count behind the conjunction.

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2. **One number per combination.** p flags: $2^p - 1$ per class.
3. **Ever rarer.** Each added flag shrinks the count behind the conjunction.

> Multiplying the parts, $0.60 \cdot 0.70 = 0.42$, is correct only if D and F are **independent given the class** — the assumption of **naive Bayes**.

The Training Table Is a Sample

amount (€)	country	hour	new device?	… 30 flags	fraud?
49.90	DE	14	no	…	no
1,250.00	BR	3	yes	…	**yes**
12.50	DE	19	no	…	no

Each column is a **random variable** X_j, each row one **draw** from $p(x, y)$. The fraud rows are draws from $p(x \mid \text{fraud})$.

Estimate $p(x \mid \text{fraud})$ by counting: of 100 frauds, how many look exactly like the next transaction?

Speaker notes

This is the reading of a table set up in Lecture 02: a column is a random variable, a row is one realisation of the vector $(X_1, \ldots, X_d, Y)$, and the table is n draws from a joint distribution nobody has written down. Restricting to the rows with $y = \text{fraud}$ is conditioning on the class, so those rows are a sample from $p(x \mid \text{fraud})$.

Estimating a distribution without any assumption about its form means counting: the estimate of $p(x \mid \text{fraud})$ for a particular vector x is the share of fraud rows equal to x in every attribute. A continuous attribute such as the amount makes exact repeats practically impossible; even after binning it, the number of value combinations grows multiplicatively with every column, as the next two slides count.

The Training Table Is a Sample

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Each column is a **random variable** X_j, each row one **draw** from $p(x, y)$. The fraud rows are draws from $p(x \mid \text{fraud})$.

Estimate $p(x \mid \text{fraud})$ by counting: of 100 frauds, how many look exactly like the next transaction?

> Almost always **none**. Counting needs examples of every combination of values.

The Joint Explodes

30
yes/no red flags per transaction

→

$2^{30} - 1$
≈ 1.07 billion numbers for $p(x \mid \text{fraud})$ alone

→

100
frauds among 100,000 transactions to estimate them

Almost every cell of the table is **empty**. Its count is 0, so its maximum likelihood estimate is 0 — every fraud that looks slightly new becomes "impossible".

Speaker notes

A joint distribution over 30 binary variables has $2^{30} = 1{,}073{,}741{,}824$ cells; since they sum to 1, $2^{30} - 1 = 1{,}073{,}741{,}823$ are free parameters, per class. With 100 fraud cases at most 100 cells have a non-zero count, so at least 99.99999% of the cells have MLE exactly 0, and any fraud pattern not seen in training receives $p(x \mid \text{fraud}) = 0$ and therefore $P(\text{fraud} \mid x) = 0$ (or an undefined $0/0$ if the pattern was not seen among honest transactions either).

A share computed from a tiny count is an extreme value, here multiplied across a billion cells; naive Bayes meets the same zero in a single cell, and Lecture 03 names the estimate behind it. More data does not solve it: filling the table reliably would need far more frauds than cells. The general pattern, and the two ways to reduce the number of parameters, follow on the next two slides.

The Curse of Dimensionality

attributes	value combinations (10 values each)	frauds per combination (100 frauds)
1	10	10
2	100	1
3	1,000	0.1
30	10^{30}	10^{-28}

Every attribute **multiplies** the cells and **divides** the data per cell. Counting needs data in every cell — a full joint over many attributes can never be estimated by counting.

Speaker notes

With p attributes of m values each, the joint has m^p cells, and n observations put on average n/m^p observations into each. Already at three attributes with ten values the average cell holds 0.1 frauds, so most cells are empty and their counts estimate 0. The name "curse of dimensionality" (Bellman) covers this exponential growth of the space with the number of dimensions; it affects every method that relies on having nearby examples, not only counting. Continuous attributes make it worse: without binning every value is its own cell.

The required amount of data grows exponentially in p, while real datasets grow by collecting more rows, not by an exponential factor per added attribute. Adding attributes therefore makes an unrestricted estimate worse, not better, unless the model is restricted.

Two Ways Out

Simplify the structure

Assume attributes are **independent given the class**: one small table per attribute instead of one huge joint.

30 flags: **30** numbers per class, not $2^{30} - 1$.

NAIVE BAYES — TODAY; GRAPHS — LECTURE 05

Use a parametric family

Replace a table of counts by a **formula with few parameters**.

amount given fraud ~ log-normal(μ, σ): **2** numbers, not 100 bins.

LECTURE 02

Both are **assumptions**. They make the model estimable — and wrong wherever the world disagrees with them.

Speaker notes

The two ways reduce the number of parameters at different places. Structure: conditional independence factorises $p(x \mid y) = \prod_j p(x_j \mid y)$, so the parameter count grows linearly in the number of attributes instead of exponentially; a Bayesian network keeps selected dependences and still factorises. Parametric families: each factor is described by a family with a fixed small number of parameters (Bernoulli for a flag, categorical for the country, normal or log-normal for an amount), so a continuous attribute needs no bins at all.

Naive Bayes uses both: independent attributes given the class, each with a small family. The resulting estimates are stable from little data, but correlated attributes are counted twice and a misfitting family distorts the likelihood ratio; both costs are taken up in the naive Bayes module.

Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

Speaker notes

Each class needs a full distribution over the $2^3 = 8$ flag patterns, which has $8 - 1 = 7$ free numbers because the probabilities sum to 1. Two classes give 14; the class prior $P(\text{fraud})$ adds one more ($P(\text{honest}) = 1 - P(\text{fraud})$ is determined). Total: 15. Typical miscounts are 16 (counting both class probabilities) and 17 (counting all 8 cells per class).

General formula for K classes and p binary attributes: $K(2^p - 1) + (K - 1)$. Under the naive assumption it becomes $Kp + (K - 1)$, for the example $2 \cdot 3 + 1 = 7$.

Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

15. Each class needs a full table over $2^3 = 8$ flag patterns: 7 free numbers (they sum to 1). Two classes give 14, and $P(\text{fraud})$ adds one.

Takeaway

The generative route models **how the data arises** — $p(y)$ and $p(x \mid y)$ — and converts with Bayes. Counting that joint needs data in every combination of values, which **explodes** with the number of attributes. The way out is an assumption: **simpler structure**, **parametric families**, or both.

Naive Bayes

Guiding Question: Its central assumption is almost always false, and it classifies well anyway. How can both be true?

Conditional Independence

Definition. A and B are **conditionally independent given** C, written $A \perp\!\!\!\perp B \mid C$, iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C) \quad \Longleftrightarrow \quad P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B tells us nothing more about A.**

Among frauds only: does a new device make a foreign IP more likely?

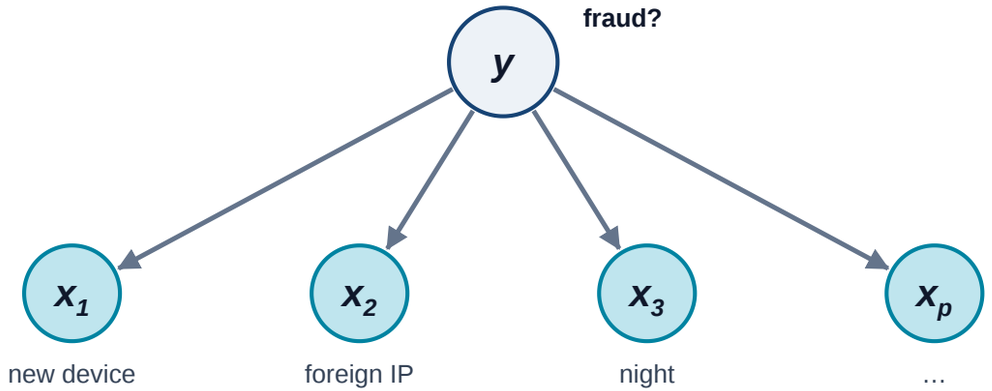
Speaker notes

The two forms are equivalent whenever $P(B \cap C) > 0$: dividing $P(A \cap B \mid C) = P(A \mid C)P(B \mid C)$ by $P(B \mid C)$ gives $P(A \mid B \cap C) = P(A \mid C)$. The relation is symmetric in A and B. For random variables it must hold for every combination of values, and the condition is evaluated separately within each value of C.

Conditional and marginal independence are different properties; neither implies the other. Two flags can be dependent overall because both are more frequent among frauds, yet independent within each class; conversely, conditioning can create a dependence between independent causes (Lecture 05 treats both cases with a counted example).

Answer to the question: in an account take-over the attacker typically uses both a new device and a foreign IP, so among frauds the two flags are positively dependent. With the figures used later in this module, $P(F \mid D, \text{fraud}) = 0.55/0.60 \approx 0.92$, compared with $P(F \mid \text{fraud}) = 0.70$. This is the dependence that naive Bayes assumes away.

The Naive Assumption



$p(y) \cdot p(x_1 \mid y) \cdot p(x_2 \mid y) \cdot \ldots \cdot p(x_p \mid y)$

Assume the attributes are **independent given the class**:

$$p(x_1, \ldots, x_p \mid y) = \prod_{j=1}^{p} p(x_j \mid y)$$

flags	full joint, per class	naive
3	7	3
30	1,073,741,823	30

Shortcut 2 — assume it apart. Dropped: every dependence *within* a class.

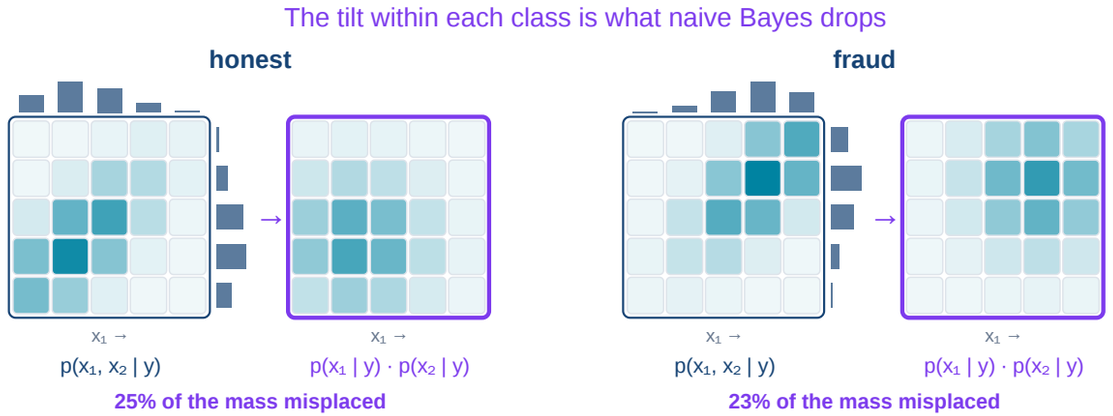
Speaker notes

In the graph the class y points to every attribute and no arrow joins two attributes. Read as a Bayesian network (Lecture 05), each missing arrow is an independence claim: every attribute is independent of all others given the class. The attributes are still dependent marginally, through the class.

Parameter count: with the assumption, $p(x \mid y)$ needs one Bernoulli parameter per binary flag and class, p numbers instead of $2^p - 1$. For a categorical attribute with k_j values the count is $k_j - 1$ per class. The three-flag example drops from 15 numbers to $2 \cdot 3 + 1 = 7$; for 30 flags from about $2.1 \cdot 10^9$ to 61.

The model is still generative, since it specifies $p(y)$ and $p(x \mid y)$ and can generate transactions. What it drops is every dependence between attributes within a class (shortcut 2 of the course roadmap). The assumption is almost never exactly true; whether it matters depends on the task, as the last slides of the module show.

Assuming It Apart, Drawn



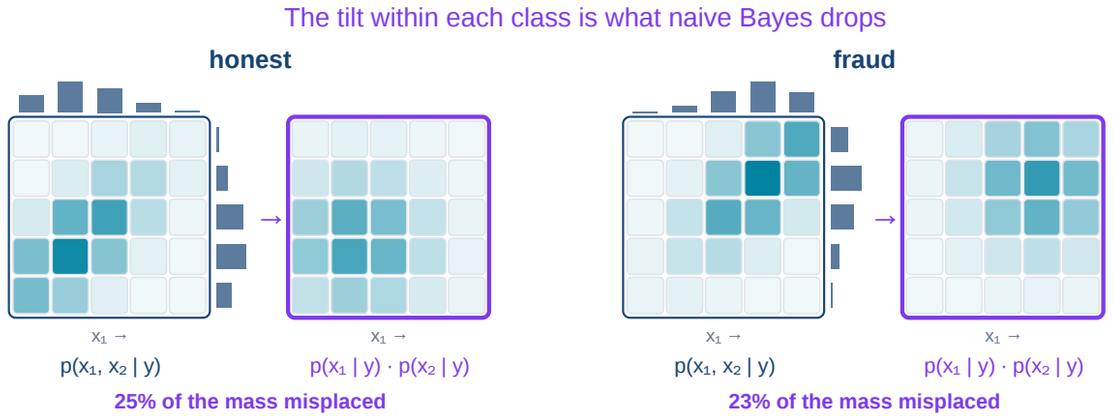
Predict: set the correlation to 0. What happens to the two pictures?

Speaker notes

The full object is a cube, class × x_1 × x_2; the figure shows its two class slices. Left: the true class-conditional joint $p(x_1, x_2 \mid y)$ on a 5×5 grid, $5^2 - 1 = 24$ numbers per class, and $5^p - 1$ for p attributes. Its two margins, $p(x_1 \mid y)$ and $p(x_2 \mid y)$, need $2 \cdot 4 = 8$. Right: the product of the margins, which is what naive Bayes stores. A product of margins is always axis-aligned — the same picture as the independence rectangle of Lecture 02 — so it cannot represent a tilt. The share of probability mass placed differently from the true joint (the total variation distance) is shown per class.

Two dependences must be kept apart. Within a class, a tilted blob means x_1 and x_2 are correlated given y; this is what naive Bayes drops. Across classes, the two blobs sit in different corners, so over all transactions x_1 and x_2 are dependent even when $\rho = 0$ — through the class, which naive Bayes keeps. Conditional independence given y does not imply independence overall.

Assuming It Apart, Drawn



Predict: set the correlation to 0. What happens to the two pictures?

They become **identical**. Naive Bayes is exact exactly when the attributes are independent **within** each class.

Fitting Is Counting

$$\hat{P}(y) = \frac{n(y)}{n} \qquad \hat{P}(x_j = v \mid y) = \frac{n(x_j = v,\ y)}{n(y)}$$

One small table per attribute

Each attribute, within one class, is a **categorical** variable — for a yes/no flag, a Bernoulli (Lecture 02).

Estimated by its share

$P(\text{new device} \mid \text{fraud})$: of the 100 frauds, how many came from a new device? 60, so 0.6.

Naive Bayes is fitted in **one pass over the table**, with no optimiser. Why a share is the right estimate — and when it is not — is the last module.

Speaker notes

Derivation. Under the naive assumption the log-likelihood of a labelled table is $\sum_i \log P(y_i) + \sum_i \sum_j \log p(x_{ij} \mid y_i)$. Grouping terms by attribute j and class y gives one term $\sum_v n(x_j = v, y) \log \theta_{jvy}$ per attribute and class, plus one term for the class prior. No parameter appears in two terms, so each is maximised separately, and each is the log-likelihood of a categorical distribution whose maximum likelihood estimate is the relative frequency (derived for two values as h/n in Lecture 03; for k values with a Lagrange multiplier). The categorical distribution was introduced in Lecture 02 as the distribution for a nominal attribute.

The independence assumption is what makes the MLE a closed-form count: with a full joint the parameters are per combination of values, and the counts would be of entire rows. Logistic regression, by contrast, has no closed-form MLE and is fitted iteratively. Every estimated number in naive Bayes is a peak of a likelihood, so shortcut 1 is taken in addition to shortcut 2; the last module makes this precise.

The Classifier

$$P(y \mid x_1, \ldots, x_p) = \frac{P(y) \prod_j p(x_j \mid y)}{\sum_{y'} P(y') \prod_j p(x_j \mid y')}$$

For two classes, divide the two posteriors — the normaliser cancels:

$$\underbrace{\frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)}}_{\text{posterior odds}} = \underbrace{\frac{P(\text{fraud})}{P(\text{honest})}}_{\text{prior odds}} \cdot \prod_{j=1}^{p} \underbrace{\frac{p(x_j \mid \text{fraud})}{p(x_j \mid \text{honest})}}_{\text{likelihood ratio } \mathrm{LR}_j}$$

> Every observed attribute **multiplies the odds** by its likelihood ratio: above 1 it speaks for fraud, below 1 against.

Speaker notes

The first line is Bayes' theorem with the naive product in place of $p(x \mid y)$; the denominator is the law of total probability. In the ratio of two posteriors the denominator cancels, which gives the odds form: posterior odds = prior odds × product of likelihood ratios. Odds and their relation to probabilities, $P = \text{odds}/(1 + \text{odds})$, were introduced with Bayes' theorem in Lecture 02.

Each attribute contributes one factor LR_j. A ratio above 1 raises the odds of fraud, below 1 lowers them, and exactly 1 means the attribute is uninformative. Because multiplication is commutative, the order in which evidence arrives does not matter. An observed absent flag also contributes a factor, $(1 - p_{j,\text{fraud}})/(1 - p_{j,\text{honest}})$, which is usually below 1 for a red flag. For more than two classes, odds relative to one reference class play the same role.

Three Red Flags

100,000 transactions, 100 of them fraud: prior odds $1 : 999$.

flag	among frauds	among honest	likelihood ratio
new device	60%	10%	6
foreign IP	70%	5%	14
night	50%	20%	2.5

A transaction shows all three flags. Is it more likely fraud than not?

Speaker notes

Arithmetic: $0.6/0.1 = 6$, $0.7/0.05 = 14$, $0.5/0.2 = 2.5$; the product is 210, and the posterior odds are $210 : 999$, so $P(\text{fraud} \mid x) = 210/(210 + 999) = 0.174$.

The answer is no: the transaction is still more likely honest. A combined likelihood ratio of 210 is strong evidence, but the prior odds of 1 : 999 are stronger; the posterior exceeds 50% only when the combined ratio exceeds 999. This is the base-rate effect of Lecture 02. Intuitive judgements that call such a transaction "clearly fraud" neglect the base rate.

The model in this example contains exactly these three flags, all observed present, so only the three ratios enter. In a model with more flags, every flag observed absent contributes its own factor; for night that factor would be $(1 - 0.5)/(1 - 0.2) = 0.625$.

Three Red Flags

100,000 transactions, 100 of them fraud: prior odds $1 : 999$.

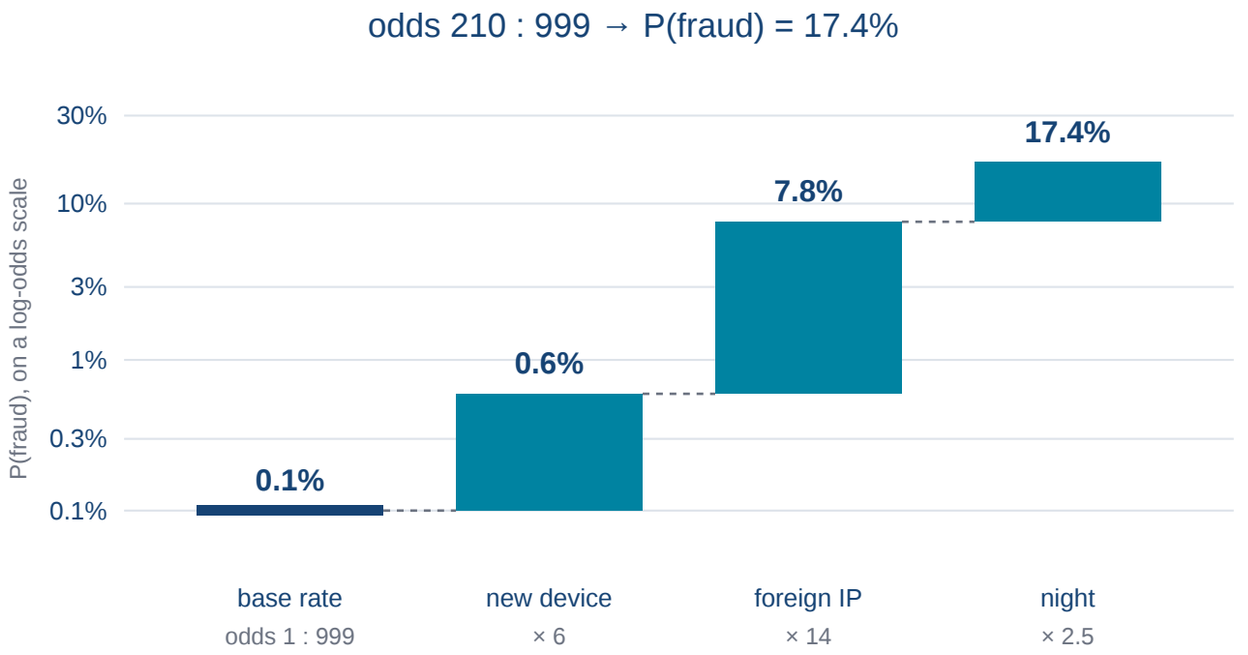
flag	among frauds	among honest	likelihood ratio
new device	60%	10%	6
foreign IP	70%	5%	14
night	50%	20%	2.5

A transaction shows all three flags. Is it more likely fraud than not?

$$\text{odds} = \frac{1}{999} \cdot 6 \cdot 14 \cdot 2.5 = \frac{210}{999} \quad \Longrightarrow \quad P(\text{fraud} \mid x) = \frac{210}{1{,}209} \approx 17\%$$

> Three red flags, each one strong — and still **mostly honest**. The base rate of Lecture 02, once more.

Evidence Adds Up



Predict before each click: which flag moves the probability most?

Speaker notes

On a log-odds axis each likelihood ratio is a step of fixed height $\log \mathrm{LR}_j$, independent of the current position: the log-odds are the sum of the log prior odds and the log likelihood ratios. Natural logarithms: prior $\log(1/999) = -6.91$; steps $\log 6 = 1.79$, $\log 14 = 2.64$, $\log 2.5 = 0.92$.

Step 1 (new device): odds $6/999$, probability 0.6%. Step 2 (foreign IP): odds $84/999$, probability 7.8%. Step 3 (night): odds $210/999$, probability 17.4%. The foreign IP has the largest ratio and therefore the largest step; night is the weakest of the three, although it moves the probability more in absolute terms than the first flag did, because probability is not linear in log-odds.

A sum of per-attribute contributions plus a constant is a linear score. For binary attributes the naive Bayes log-odds are exactly linear in x, the same functional form as logistic regression (Lecture 04); the two differ in how the weights are estimated (Lecture 04).

One Zero Vetoes Everything

A new flag: **payment to a crypto exchange**. Among the 100 frauds it never occurred.

$$\hat{P}(\text{crypto} \mid \text{fraud}) = \frac{0}{100} = 0$$

A transaction with this flag gets posterior odds **exactly 0** — whatever the other thirty flags say. One unseen value vetoes all other evidence.

COUNTING, FAILS

"Not seen in 100 frauds" became "impossible for a fraud" — a share of a small count, read as certainty.

Speaker notes

This is the zero-frequency problem. The MLE of a count of 0 is the probability 0, and a single factor of 0 makes the whole product 0, so the posterior odds of fraud are 0 regardless of all other likelihood ratios. If a value has count 0 in both classes, the posterior is the undefined $0/0$.

The cause is that a relative frequency of a small count is an extreme value; Lecture 03 shows the same failure on a coin (three heads in three flips give $\hat{\theta} = 1$) and names the estimate. With 100 frauds, any flag with a true rate below about 1% is likely to have count 0 (for a true rate of 1% the probability of no occurrence in 100 is $0.99^{100} = 0.37$). In a log-odds implementation the zero appears as $\log 0 = -\infty$. Standard repairs add pseudo-counts (next slide) or replace zeros by a small floor value; the first has a probabilistic justification.

Laplace Smoothing

Laplace smoothing pretends each of the k_j values was seen **once more**:

$$\hat{P}(x_j = v \mid y) = \frac{n(x_j = v,\ y) + 1}{n(y) + k_j}$$

For a yes/no flag ($k_j = 2$) this is $\frac{h+1}{n+2}$.

count among 100 frauds	share	smoothed
0	0	$1/102 \approx 0.010$
60	0.60	$61/102 \approx 0.598$

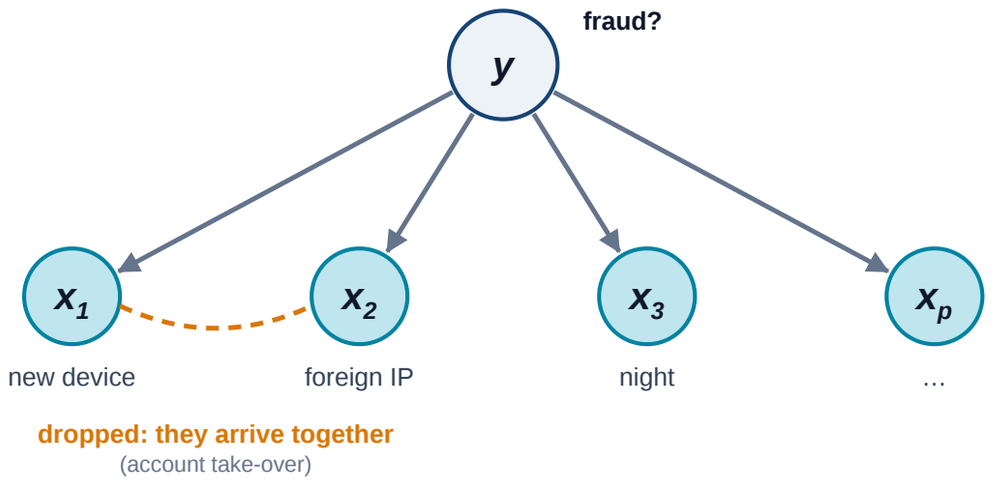
> Where the count is large, nothing changes; where it is 0, the veto is gone. Why **+1**, and not +5? The next module: smoothing is a **prior**.

Speaker notes

The pseudo-counts are added to every value of the attribute, so the smoothed estimates still sum to 1 over the k_j values: $n(y) + k_j$ in the denominator balances one extra count per value. On well-populated cells the change is negligible (0.60 vs 0.598); on empty cells it replaces 0 by $1/102 \approx 0.0098$, so no single factor can force the product to 0.

Add-α smoothing, $(h + \alpha)/(n + 2\alpha)$ for a flag, generalises the rule to α imagined occurrences per value; larger α pulls every estimate more strongly towards one half. The choice $\alpha = 1$ is a convention (Laplace's rule of succession). The next module derives it: $(h + 1)/(n + 2)$ is the posterior mean of the flag probability under a flat Beta$(1, 1)$ prior (Lecture 03), and add-α is the posterior mean under Beta(α, α); for $k_j > 2$ values the flat Dirichlet plays the same role.

Counting Evidence Twice



In an account take-over, a new device and a foreign IP **arrive together**. Suppose both appear on 55% of frauds and 4% of honest transactions:

	ratio for the pair	$P(\text{fraud})$ with night
naive Bayes	$6 \cdot 14 = 84$	17.4%
the real pair	$0.55/0.04 \approx 14$	3.3%

Correlated flags are **counted twice**, so naive Bayes is **over-confident**. Its ranking of transactions often survives; its probabilities do not.

Speaker notes

The dashed edge in the figure is the dependence between new device and foreign IP that naive Bayes omits. Naive Bayes multiplies both likelihood ratios as if they were independent pieces of evidence, $6 \cdot 14 = 84$. The pair's actual likelihood ratio is $P(D, F \mid \text{fraud}) / P(D, F \mid \text{honest}) = 0.55 / 0.04 = 13.75$. With night added, the posterior odds are $13.75 \cdot 2.5 / 999 = 34.4 / 999$, so $P(\text{fraud} \mid x) = 3.3\%$ instead of 17.4%. Under independence within the classes the pair would occur on $0.6 \cdot 0.7 = 42\%$ of frauds and $0.1 \cdot 0.05 = 0.5\%$ of honest transactions; the assumed 4% among honest ones is eight times that, which is where the error comes from.

Correlated attributes therefore push naive Bayes posteriors towards 0 and 1: the probabilities are not calibrated (Lecture 04). Classification by arg max depends only on which side of the decision threshold the posterior falls, so it can remain correct even when the probabilities are badly off. Domingos and Pazzani (1997, Machine Learning 29) show that naive Bayes can be optimal under zero–one loss in many cases where the independence assumption is violated, which is why it classifies well despite a false assumption.

Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

Speaker notes

Answer: up. The duplicated flag contributes its likelihood ratio twice, so the odds become $210 \cdot 14/999 = 2940/999$ and $P(\text{fraud} \mid x) = 2940/3939 = 0.746$, from 17%.

The true information content is unchanged, since a copy of an attribute carries no new evidence: the correct likelihood ratio of the pair (flag, copy) is just the ratio of the flag, 14. Naive Bayes cannot detect duplication, because the model has no parameter for the relation between attributes. A duplicate is the limiting case of perfect correlation; strongly correlated attributes act as partial copies. Real feature sets contain them routinely: the same information coded twice, features derived from others, and indicators that are near-functions of each other. Removing redundant attributes, or a model that captures the dependence (a Bayesian network, Lecture 05), avoids the double counting.

Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

Up — from 17% to about 75%. The odds get a second factor 14: $2{,}940 : 999$. Naive Bayes cannot tell a copy from new evidence; a strongly correlated attribute is almost a copy.

Takeaway

Naive Bayes **assumes the joint apart** — attributes independent given the class — and fits every factor by **counting**, smoothed by pseudo-counts so no zero can veto. It is cheap, needs little data and ranks well. Its probabilities are over-confident wherever attributes move together.

Fitting the Parameters: Posterior, MLE and MAP

Guiding Question: Naive Bayes fitted itself by counting, and a count of zero vetoed everything. What was counting, really — and what does a prior change?

An Unknown Number, Ten Observations

A new feature went live. Of the first **10 users**, **3 came back**.

What is p, the probability that a user comes back — and how sure are we?

Speaker notes

The frequentist numbers: $\hat{p} = 3/10 = 0.3$, $\mathrm{SE} = \sqrt{0.3 \cdot 0.7/10} \approx 0.145$, and the normal-approximation (Wald) interval $0.3 \pm 1.96 \cdot 0.145 = [0.016, 0.584]$. The normal approximation is poor at $n = 10$: the binomial is discrete and, for p away from 0.5, skewed, and the Wald interval is known to cover less often than its nominal 95% for small n. With 0 of 10 it would even give the degenerate interval $[0, 0]$.

The confidence interval forbids the sentence "p lies in this interval with probability 0.95", because p is treated as a fixed number. The Bayesian alternative describes the uncertainty about p with a probability distribution, as for any other unknown quantity; the sentence then becomes a valid statement about that distribution.

An Unknown Number, Ten Observations

A new feature went live. Of the first **10 users**, **3 came back**.

What is p, the probability that a user comes back — and how sure are we?

WHAT THE LAST UNIT CAN SAY

$\hat{p} = 0.3$, standard error ≈ 0.14, a 95% interval of roughly $[0.02, 0.58]$ — about the *procedure*, not about p.

WHAT WE WOULD LIKE TO SAY

"p is probably between 0.1 and 0.6, most likely around 0.3" — a statement **about** p.

Conditioning on the Data — for a Parameter

Treat the unknown p as a random variable θ and apply **Bayes' theorem**, with θ in place of the hypothesis and the data D as the evidence:

$$p(\theta \mid D) = \frac{p(D \mid \theta) \cdot p(\theta)}{p(D)}$$

posterior — belief about θ after the data

likelihood — how probable the data is for each θ

evidence — normalises, does not depend on θ

prior — belief about θ before

For i.i.d. yes/no data with h yeses in n trials: $p(D \mid \theta) = \theta^h(1-\theta)^{n-h}$ — read as a function of θ, with the data fixed.

Speaker notes

The structure is that of Bayes' theorem for events, with the hypothesis replaced by a number $\theta \in [0, 1]$: prior and posterior are densities over $[0, 1]$ instead of two probabilities. The evidence is $p(D) = \int_0^1 p(D \mid \theta) p(\theta) d\theta$, the marginal probability of the data; it does not depend on θ and only normalises.

The likelihood is the Bernoulli PMF multiplied over the n rows, which is where the i.i.d. assumption enters. Only the counts h and $n - h$ matter, not the order of the observations. The binomial coefficient is absent because D is one specific sequence; including it would change nothing, since it cancels in the normalisation. As a function of θ the likelihood is not a probability distribution: $\int_0^1 \theta^h (1 - \theta)^{n-h} d\theta$ is not 1 in general. Lecture 03 builds on this distinction.

A Prior That Behaves Like Counts

Beta prior. $p(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}$ on $[0,1]$, written $\mathrm{Beta}(a,b)$. $\mathrm{Beta}(1,1)$ is flat: every p equally plausible.

Multiply by the likelihood $\theta^h(1-\theta)^{n-h}$:

$$p(\theta \mid D) \propto \theta^{a+h-1}(1-\theta)^{b+n-h-1} \quad \Longrightarrow \quad \theta \mid D \sim \mathrm{Beta}(a+h,\ \ b+n-h)$$

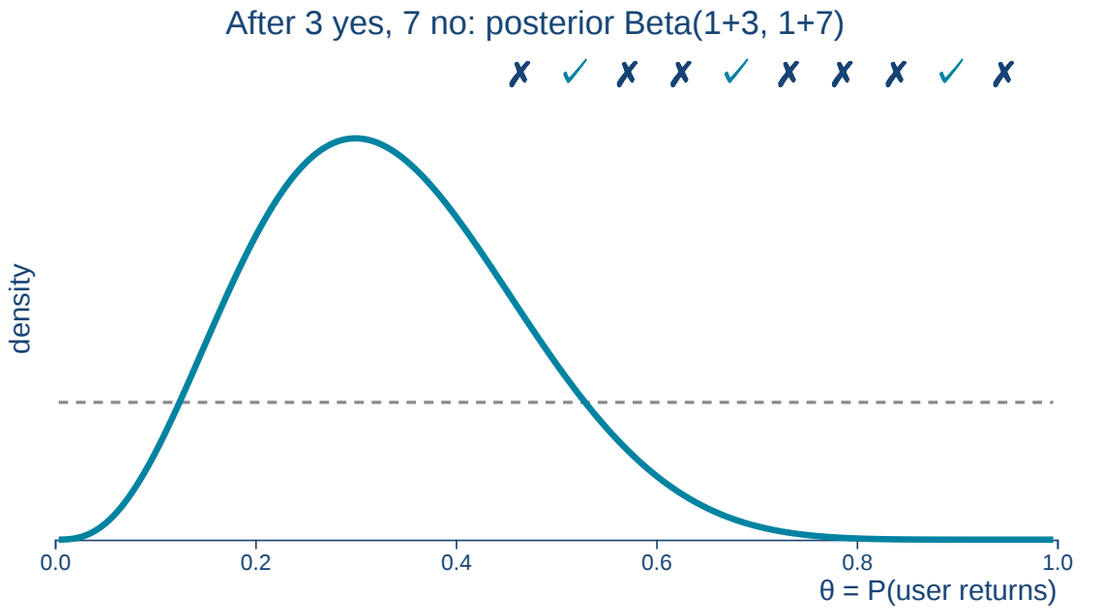
The update is **adding counts**: a and b act like yeses and noes seen before the data. A prior whose posterior stays in the same family is called **conjugate**.

Speaker notes

The normalising constant of $\mathrm{Beta}(a, b)$ is $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$, but it never has to be computed for the update: prior and likelihood are both of the form $\theta^{\cdot}(1 - \theta)^{\cdot}$, so their product is again of that form and the exponents add. The posterior is identified by its shape alone.

The parameters act as pseudo-counts: $\mathrm{Beta}(a, b)$ behaves like $a - 1$ yeses and $b - 1$ noes already seen in terms of the peak, or a and b in terms of the mean $a/(a + b)$. A prior with $a + b = 2$ is outweighed by 1,000 real observations. Conjugate priors exist for other likelihoods too, e.g. the Dirichlet for categorical data and the normal for the mean of normal data with known variance.

Watching the Posterior Form



Predict before each click: which way does the curve move after a ✗ — and does it get narrower every time?

Speaker notes

Step 0 shows the flat prior; each further step adds one observation of the sequence ✗✓✗✗✓✗✗✗✓✗. After the first ✗ the posterior is $\mathrm{Beta}(1, 2)$, density $2(1 - \theta)$, highest at $\theta = 0$ and ruling out only $\theta = 1$. After the first ✓ it is $\mathrm{Beta}(2, 2)$, peaked at 0.5. Every ✗ moves the peak left and every ✓ moves it right. After all ten: $\mathrm{Beta}(4, 8)$, peak 0.3, mean 0.33.

The curve does **not** narrow at every step. Its standard deviation falls from 0.289 (flat) to 0.138 after nine observations, but the ninth observation, a ✓ after five ✗ in the last six, raises it slightly to 0.139, because the data then look less one-sided. In general an observation that contradicts the current belief can widen the posterior; only on average over possible observations does the posterior variance decrease. The overall narrowing is the epistemic uncertainty shrinking, as described in Lecture 01.

What the Posterior Gives You

After 3 of 10, with a flat prior: $\theta \mid D \sim \mathrm{Beta}(4, 8)$.

question	answer	from
most plausible p	$0.30 = h/n$	the peak (mode)
best guess on average	0.33	the posterior mean $\frac{a+h}{a+b+n}$
where is p, 95%?	$[0.11,\ 0.61]$	a **credible interval**
after 1,000 users, 300 back	$[0.27,\ 0.33]$	same model, more data

> The peak under a flat prior is exactly h/n. **Lecture 03** keeps only this peak and calls it maximum likelihood — the first shortcut of the course.

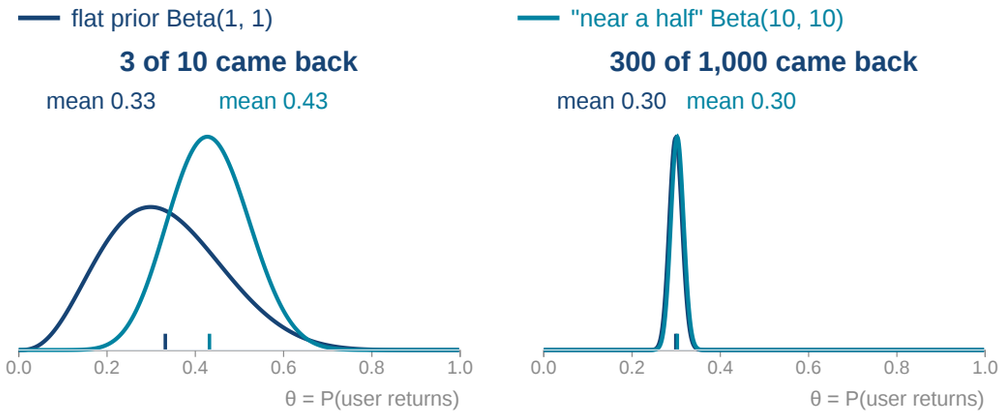
Speaker notes

All rows follow from $\mathrm{Beta}(4, 8)$. Mode: $(a-1)/(a+b-2) = 3/10$. Mean: $4/12 \approx 0.33$; it lies to the right of the mode because the posterior is right-skewed. The 95% interval is the equal-tailed one, from the 2.5% and 97.5% quantiles: $[0.109, 0.610]$. It supports the statement "p lies in $[0.11, 0.61]$ with probability 0.95", given the prior and the model. A highest-density interval, the shortest interval with 95% probability, is a common alternative and differs slightly for skewed posteriors.

With 300 of 1,000 the posterior is $\mathrm{Beta}(301, 701)$, interval $[0.272, 0.329]$. The posterior standard deviation falls from 0.131 to 0.0145, a factor of about 9 for 100 times the data, close to the $1/\sqrt{n}$ law. Under a flat prior the posterior is proportional to the likelihood, so the mode equals the maximum-likelihood estimate h/n.

How Much Does the Prior Matter?

Two analysts: one with a flat prior $\mathrm{Beta}(1,1)$, one convinced that p is near a half, $\mathrm{Beta}(10,10)$.



> With little data the prior matters; with enough data the analysts **agree**. A prior is a stated assumption — it can be argued with, and it washes out.

Speaker notes

Step 0 shows the posteriors after 3 of 10, step 1 adds 300 of 1,000; each panel has its own height scale. "Near a half" prior: $\mathrm{Beta}(10+3, 10+7) = \mathrm{Beta}(13, 17)$, mean $13/30 \approx 0.43$; with 300 of 1,000, $\mathrm{Beta}(310, 710)$, mean $310/1020 \approx 0.304$. Flat prior: means $4/12 \approx 0.33$ and $301/1002 \approx 0.300$.

The posterior mean is a weighted average of prior mean and observed proportion: $\frac{a+h}{a+b+n} = \frac{a+b}{a+b+n} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \frac{h}{n}$. The prior's weight is $(a+b)/(a+b+n)$: for $\mathrm{Beta}(10, 10)$ it is $20/30$ with 10 observations and $20/1020 \approx 2\%$ with 1,000. The prior is an explicit modelling assumption that can be criticised like any other. Where data are scarce its influence is large, and it is then the only additional information available. A prior that assigns zero density to some region can never be corrected there by data.

Mini-Exercise

Prior $\mathrm{Beta}(2, 2)$. Of 10 users, **8** come back.

Is the posterior mean above or below 0.8 — and why?

Speaker notes

The posterior is $\mathrm{Beta}(10, 4)$ with mean $10/14 \approx 0.714$ and mode $9/12 = 0.75$, both below the observed 0.8. The weighted-average form gives the reason: the mean is $\frac{4}{14} \cdot 0.5 + \frac{10}{14} \cdot 0.8 \approx 0.714$, a compromise between the prior mean 0.5 (weight $4/14$) and the data (weight $10/14$). With 80 of 100 the mean would be $82/104 \approx 0.79$.

Pulling an estimate towards a default value is called shrinkage. The same effect appears as regularisation in model fitting: a prior centred on zero weights corresponds to a penalty on large weights, which is how Lecture 04 reads weight decay.

Mini-Exercise

Prior $\mathrm{Beta}(2, 2)$. Of 10 users, **8** come back.

Is the posterior mean above or below 0.8 — and why?

Below. The posterior is $\mathrm{Beta}(2 + 8,\ 2 + 2) = \mathrm{Beta}(10, 4)$, mean $10/14 \approx 0.71$. The prior counts two yeses and two noes, so it pulls towards 0.5 — and the pull fades as data grows.

Two Ways to Answer "Where Is p?"

	random	the 95% interval	needs
Frequentist	the data; p is fixed	a **procedure** that catches p in 95% of samples — about this one interval, no probability statement	nothing but the sampling model
Bayesian	p; the data is fixed	"p lies in this interval with probability 0.95" — a true sentence about p	a **prior**: what we believed about p before the data

> The axioms are the same; the question is not. The rest of the course mostly takes the second road — and Lecture 03 starts with its cheapest version: keep only the peak.

Speaker notes

Both schools use the same probability axioms but assign probabilities to different things. Frequentist statistics assigns them only to repeatable random events, here the data; a parameter is fixed, so statements about it are statements about procedures (coverage, error rates). Bayesian statistics assigns probabilities to any uncertain quantity, including a parameter, and conditions on the observed data. The confidence interval needs no prior and makes the weaker, procedural statement; the credible interval makes the direct statement about p and requires a prior.

With a flat prior and much data the two nearly coincide numerically: for 300 of 1,000 the credible interval is $[0.272, 0.329]$ and the Wald interval $0.3 \pm 1.96 \cdot 0.0145 = [0.272, 0.328]$. This follows from the Bernstein–von Mises theorem: for large n the posterior is approximately normal around the maximum-likelihood estimate with the same spread as its sampling distribution. For small n, or an informative prior, the two intervals differ, as for 3 of 10.

Takeaway

An unknown parameter is just another unknown: give it a **prior**, condition on the data, and the **posterior** says what the data justifies — as a peak, a mean and an interval. For yes/no data with a Beta prior the update is **adding counts**, and the prior's influence fades as data grows.

What Was Counting, Really?

Naive Bayes set $P(\text{new device} \mid \text{fraud}) = 60/100$ and $P(\text{crypto} \mid \text{fraud}) = 0/100$. Which principle produces these numbers — and which one would have avoided the 0?

Speaker notes

A flag within one class is a Bernoulli variable with unknown parameter $\theta = P(\text{flag} \mid y)$. For h occurrences among n cases of the class, the probability of the observed data is $\theta^h(1-\theta)^{n-h}$, largest at $\theta = h/n$: the share. For $h = 0$ the maximum is at $\theta = 0$, which is the zero-frequency problem. Laplace smoothing replaced the share by $(h+1)/(n+2)$, the mean of the posterior under a flat prior. Both readings are taken on the returning-users example of Lecture 03, where the whole posterior is known and can be plotted.

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The share is the parameter value under which the observed data are **most probable**: the **maximum likelihood estimate**. Smoothing used a prior and took the posterior **mean**.

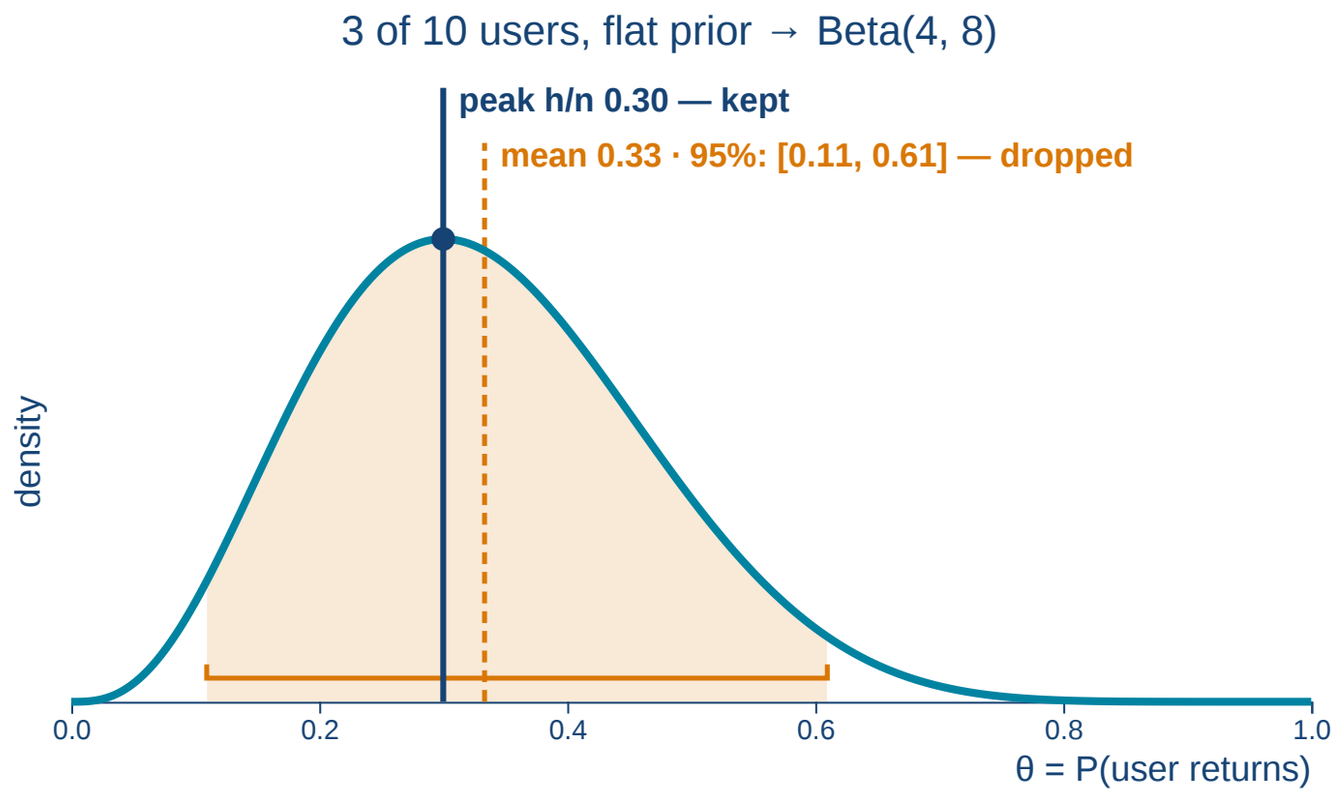
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The share is the parameter value under which the observed data are **most probable**: the **maximum likelihood estimate**. Smoothing used a prior and took the posterior **mean**.

> Both are readings of the posterior over the parameter just built — one its peak, one its mean.

One Posterior, Three Readings



- **peak** $0.30 = h/n$ — the share
- **mean** 0.33 — the smoothed value
- **95% interval** $[0.11, 0.61]$

> Naive Bayes kept only the **peak**. What does that drop — and why does almost everyone do it?

Speaker notes

The posterior of Lecture 03: 3 of 10 users return, flat prior Beta(1, 1), so the posterior is Beta(1 + 3, 1 + 7) = Beta(4, 8). Its three summaries: the mode $(a-1)/(a+b-2) = 3/10 = 0.30$, which equals h/n because the prior is flat; the mean $a/(a+b) = 4/12 = 0.33$, which is Laplace's $(h+1)/(n+2)$; the central 95% interval $[0.109, 0.610]$ from the Beta quantiles. The standard deviation is 0.13.

The peak (navy in the plot) is the only one of the three that can be found without the normaliser, by maximising; mean and interval (amber) need the whole distribution. Fitting by least squares or by cross-entropy is exactly this kind of peak: least squares is the MLE under Gaussian noise, cross-entropy is the negative log-likelihood of a Bernoulli or categorical output, and adding weight decay turns either into a MAP estimate (Lecture 04).

The Likelihood, Read Along the Other Axis

The **likelihood** is the probability of the observed data, read as a function of the **parameter** with the data held fixed: $L(\theta) = p(D \mid \theta)$. The **maximum likelihood estimate** is its peak, $\hat{\theta}_{\mathrm{ML}} = \arg\max_\theta p(D \mid \theta)$.

Lecture 03, 3 of 10 users

$$p(\theta \mid D) \propto \theta^3 (1-\theta)^7 \cdot 1$$

With a **flat prior** the posterior *is* the normalised likelihood — so its peak $h/n = 0.3$ is the maximum likelihood estimate.

Not a distribution over θ

$$\int_0^1 \theta^3 (1-\theta)^7 \mathrm{d}\theta = \frac{1}{1320}$$

The likelihood is a probability over **data**. Read along θ it need not integrate to 1 — only the posterior does.

Speaker notes

With a flat prior $p(\theta) = 1$ on $[0, 1]$, Bayes' theorem gives $p(\theta \mid D) = p(D \mid \theta)/p(D)$: the posterior is the likelihood divided by a constant, so both have the same shape and the same arg max. The MLE is therefore the MAP under a flat prior. With any other prior the two peaks differ.

The normaliser here is $\int_0^1 \theta^3(1-\theta)^7\, d\theta = B(4, 8) = 3!\,7!/11! = 1/1320$, the Beta function that turns the likelihood into the Beta(4, 8) density. That the integral is not 1 shows that the likelihood is not a density over θ: it is a probability of the data for each fixed θ, and summed over all possible datasets it gives 1, not integrated over θ. A common error is to read "the likelihood of $\theta = 0.3$" as "the probability that $\theta = 0.3$"; only the posterior supports the second statement, and only after a prior has been chosen. The likelihood is also defined only up to a constant factor in θ (for example the binomial coefficient $\binom{10}{3}$ when only the count is recorded), which changes neither its peak nor the posterior.

Deriving h/n Without a Posterior

The i.i.d. assumption of Lecture 02 lets the likelihood factorise over rows; the logarithm turns the product into a sum without moving the peak:

$$\ell(\theta) = \log \prod_{i=1}^{n} \theta^{y_i} (1-\theta)^{1-y_i} = h \log \theta + (n-h) \log(1-\theta)$$

$$\frac{d\ell}{d\theta} = \frac{h}{\theta} - \frac{n-h}{1-\theta} = 0 \quad \Longrightarrow \quad \hat{\theta}_{\mathrm{ML}} = \frac{h}{n}$$

> Last week's peak, found by **setting a derivative to zero —** no prior, no normaliser, no integral. The recipe works unchanged for a million parameters, where nobody can draw the posterior.

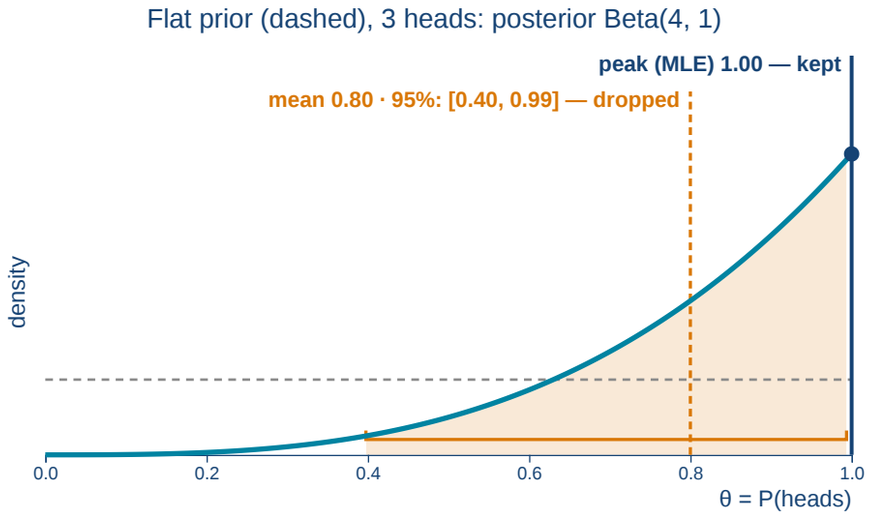
Speaker notes

Steps of the derivation. Under the i.i.d. assumption $p(D \mid \theta) = \prod_i \theta^{y_i}(1-\theta)^{1-y_i} = \theta^h(1-\theta)^{n-h}$ with $h = \sum_i y_i$, so the data enter only through the count h (a sufficient statistic). Taking the logarithm is allowed because $\log$ is strictly increasing, so it does not move the arg max; it also avoids numerical underflow of a product of many small probabilities and makes the derivative a sum of per-row terms. Solving $h/\theta = (n-h)/(1-\theta)$ gives $h(1-\theta) = (n-h)\theta$, i.e. $\theta = h/n$. The second derivative $-h/\theta^2 - (n-h)/(1-\theta)^2$ is negative, so this is a maximum. For $h = 0$ or $h = n$ the maximum lies on the boundary ($\theta = 0$ or 1), where the derivative is not zero; the formula h/n still holds.

The reason the peak is used in practice: it replaces integration (the normaliser $p(D)$) by optimisation. For models with many parameters there is no closed form, and the maximum is found numerically, typically by gradient methods on the negative log-likelihood.

The Peak Can Lie

Three flips, **three heads**. Maximum likelihood: $\hat{\theta}_{\mathrm{ML}} = 3/3 = 1$.



Predict before the second click: where do the posterior mean and the 95% interval lie?

Speaker notes

Step 1 marks the peak of Beta(4, 1) at $\theta = 1$: the MLE $3/3$ assigns probability 0 to tails, a claim no finite number of heads can justify. Step 2 adds the posterior mean and the central 95% interval. Beta(4, 1) has density $4\theta^3$ and CDF θ^4, so the mean is $4/5 = 0.8$ and the 2.5% and 97.5% quantiles are $0.025^{1/4} = 0.398$ and $0.975^{1/4} = 0.994$. The posterior therefore says "probably above one half, possibly well below 1", whereas the peak says "never tails".

The data and the model are not at fault; the information loss comes from reducing a skewed posterior to its mode. For a posterior with its mode on the boundary of the parameter space, the mode is an extreme point of the distribution and a poor summary. This is the zero-frequency problem of naive Bayes on a coin: there a count of 0 gave an estimated probability of exactly 0.

A Prior's Peak: MAP

Keep the prior, and still take only the peak — the **maximum a posteriori** estimate. With a $\mathrm{Beta}(a, b)$ prior, the same derivative gives

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} p(D \mid \theta) p(\theta) = \frac{h + a - 1}{n + a + b - 2}$$

prior	data	MLE h/n	MAP	posterior mean
$\mathrm{Beta}(1, 1)$ flat	3 of 3	1	1	0.80
$\mathrm{Beta}(2, 2)$	3 of 3	1	**0.80**	0.71
$\mathrm{Beta}(2, 2)$	8 of 10 (Lecture 03)	0.80	**0.75**	0.71

The prior shifts the peak as if $a - 1$ yeses and $b - 1$ noes had been seen before the data — a **flat** prior shifts nothing. MAP pulls 3 of 3 back to 0.8 — and still reports one number.

Speaker notes

Derivation: the log-posterior is, up to a constant, $(h + a - 1)\log\theta + (n - h + b - 1)\log(1 - \theta)$, the log-likelihood of a dataset with $a - 1$ extra successes and $b - 1$ extra failures. Its maximum is therefore $(h + a - 1)/(n + a + b - 2)$, valid when both exponents $h + a - 1$ and $n - h + b - 1$ are non-negative and not both zero (for $a, b < 1$ and few data the posterior density is unbounded at the boundary). Beta(1, 1) adds nothing and MAP equals MLE.

Rows of the table. Beta(2, 2) with 3 of 3 gives posterior Beta(5, 2): mode $4/5 = 0.80$, mean $5/7 = 0.71$. Beta(2, 2) with 8 of 10 gives Beta(10, 4): mode $9/12 = 0.75$, mean $10/14 = 0.71$; this is the mini-exercise of Lecture 03.

The posterior mean $(a + h)/(a + b + n)$ is a different point from the MAP $(a + h - 1)/(a + b + n - 2)$; they coincide only for a symmetric posterior ($a + h = b + n - h$). MAP is not invariant under reparametrisation (the mode of a density changes when θ is transformed), the MLE is. Penalised fitting of weights, such as ridge regression or weight decay, is MAP with a Gaussian prior (Lecture 04).

Why Everyone Keeps the Peak

	the whole posterior	only the peak
needs	the normaliser $p(D)$ — an integral over θ	a derivative
in 1 dimension	a Beta table	h/n
in a million	an integral nobody can compute exactly	gradient descent
costs	computation	the **spread**

With **1,000 users** the posterior is $\mathrm{Beta}(301, 701)$: standard deviation 0.015. The peak is nearly everything there is to know.

Speaker notes

The full posterior needs $p(D) = \int p(D \mid \theta) p(\theta)\, d\theta$. In one dimension with a conjugate prior this is a known Beta function; in general it is an integral over all parameters, which has no closed form for most models and grows intractable with dimension. The peak needs only the unnormalised log-posterior and its gradient.

Numbers for the note: Beta(301, 701) has standard deviation $\sqrt{ab/((a+b)^2(a+b+1))} = 0.0145$ and central 95% interval $[0.272, 0.329]$ (Lecture 03). With n observations the posterior standard deviation shrinks roughly as $\sqrt{\theta(1-\theta)/n}$, so with enough data the posterior concentrates around the peak and the peak is an adequate summary.

The approximation fails where the data say little: small samples, many parameters relative to data, and predictions far from the observed inputs. The last case persists even with large datasets; Bayesian linear regression (Lecture 07) shows predictive uncertainty growing away from the data, which a point estimate of the weights cannot express.

Mini-Exercise

Two analysts both report $\hat{\theta} = 0.3$. One saw **3 of 10** users return, the other **300 of 1,000**.

What does the reported number hide — and which estimate would you bet on?

Speaker notes

Answer: the point estimate hides the spread. 3 of 10 with a flat prior gives Beta(4, 8), 95% interval $[0.11, 0.61]$, standard deviation 0.13; 300 of 1,000 gives Beta(301, 701), interval $[0.27, 0.33]$, standard deviation 0.015. The interval widths are 0.50 and 0.057, a ratio of about 9, close to the factor $\sqrt{1000/10} = 10$ expected from a spread shrinking as $1/\sqrt{n}$.

The two estimates agree in value but not in reliability, so the 1,000-user estimate is the one to rely on. The number 0.3 alone cannot distinguish the two cases; reporting n or an interval alongside the estimate restores the missing information.

Mini-Exercise

Two analysts both report $\hat{\theta} = 0.3$. One saw **3 of 10** users return, the other **300 of 1,000**.

What does the reported number hide — and which estimate would you bet on?

The **spread**. The posteriors of Lecture 03: $[0.11, 0.61]$ after 10 users, $[0.27, 0.33]$ after 1,000 — the same peak, an interval nine times wider. Bet on the second; a point estimate cannot tell you to.

Three Choices, Made Separately

choice	options	decides
what to model	generative: $p(y)$, $p(x \mid y)$ · discriminative: $p(y \mid x)$	which distribution is learned
how to fit θ	MLE · MAP · keep the whole posterior	which parameters we keep
how to decide	most probable class · a threshold from the costs	what we do with $P(y \mid x)$

Two different "MAP"s. MAP *estimation* picks the most probable **parameter** given the training data. The MAP *decision* picks the most probable **class** for one input, $\arg\max_y P(y \mid x)$.

Speaker notes

The three choices can be combined freely. Naive Bayes is generative; fitted by counting it is the MLE, with Laplace smoothing the posterior mean under a flat prior, and with a Beta(a, b) prior and the mode it is MAP. Logistic regression (Lecture 04) is discriminative; fitted by minimising cross-entropy it is the MLE, with an L2 penalty the MAP estimate under a Gaussian prior. Either model can keep its whole posterior over parameters instead (Bayesian logistic regression, Bayesian naive Bayes), at the cost of an integral.

The decision is a third, separate step. The MAP decision rule $\arg\max_y P(y \mid x)$ minimises the error rate; with unequal costs of errors the threshold moves (Lecture 06). The "maximum likelihood decision" $\arg\max_y p(x \mid y)$ ignores the class prior; for rare classes it is the base-rate error of Lecture 02. The name clash between MAP estimation and the MAP decision is common in textbooks; both maximise a posterior, but over different unknowns.

Takeaway

Shortcut 1 — keep the peak. Maximum likelihood (flat prior) and MAP (with a prior) keep the posterior's most probable point; naive Bayes' counting was the first. It is cheap — a derivative instead of an integral — and harmless with plenty of data. It drops the **spread**, and it is a choice of its own: generative or discriminative, every model is fitted this way.